

HACKATHON SQUAD



Your university is hosting its flagship annual hackathon, and you want to assemble the ultimate dream team from a massive pool of freshman coders. Each coder has a specific "Skill Rating" based on their coding speed and problem-solving abilities, which contributes to the team's overall strength.

However, due to past rivalries and department drama, certain pairs of coders refuse to work together. As the team captain, your task is to design an algorithm to select a subset of coders that maximizes the total skill rating of the squad without including any conflicting pairs.

Due to the massive scale of the student body, finding the absolute perfect mathematical team might be computationally impossible. Your goal is to apply optimization techniques to find a valid team with the **highest possible** total skill rating within the time limit.

GOALS

- Total Score: Sum of the skill ratings of all selected coders.
- Conflict-Free Constraint: The selected team must be strictly conflict-free; no two coders in the selected subset can have a predefined rivalry.
- Team Size: A team can consist of any number of coders, from just one individual to the entire available pool.
- Objective: Design and implement a heuristic or approximation algorithm that rapidly evaluates valid combinations and outputs the highest-scoring subset it can find within the 5-minute execution window.

INPUT FORMAT

- The first line contains two integers N and M — the total number of available coders and the number of conflict pairs.
- The second line contains N space-separated integers: $S_1, S_2, \dots, S_n$, where S_i represents the skill rating of the i -th coder.
- The next M lines each contain two integers: u and v , indicating a mutual conflict between coder u and coder v .

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OUTPUT FORMAT

- The first line must contain a single integer — the total sum of skill ratings for your selected, conflict-free team.
- The second line should list the space-separated indices of the selected coders in ascending order to prove the team composition.
- (Note: Your solution will be judged based on the validity of your team and how high your score is relative to the best known solutions.)

CONSTRAINTS

- $1 \leq N \leq 200,000$
- $0 \leq M \leq (N * (N - 1)) / 2$
- $1 \leq S_i \leq 1,000,000,000$
- $1 \leq u, v \leq N$
- $u \neq v$
- All conflict pairs (u, v) are distinct.